Regression Modeling Project: Project 1

Predicting Baccalaureate Graduation Rates

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DSC 531 | Spring 2026

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1. Introduction

This report details the development of a linear regression model to predict baccalaureate graduation rates, defined as the proportion of students completing their degree within 150% of normal time (i.e., six years for a four-year program), using data from the Integrated Postsecondary Education Data System (IPEDS). The analysis is restricted to institutions with an incoming cohort of at least 200 students, yielding an analytic sample of 1,252 institutions prior to merging with additional data sources.

2. Data

2.1 Data Sources

Four IPEDS datasets were merged using the institution identifier (UNITID) as the common key:

- graduation.sas7bdat: graduation rates and incoming cohort counts
- characteristics.sas7bdat: institutional characteristics (control, locale, Carnegie classification, etc.)
- tuitionandcosts.sas7bdat: tuition, fees, and housing information
- salaries.sas7bdat: faculty salary information (filtered to all instructional staff total, rank code 7)

The graduation dataset contained two rows per institution: one for the incoming cohort and one for completers within 150% of normal time. The outcome variable, gradrate, was computed as the ratio of completers to incoming cohort size. An inner join across all four datasets was used to ensure complete data, resulting in a final analytic dataset of 1,252 institutions.

2.2 Outcome Variable

The outcome variable gradrate is a continuous proportion ranging from 0.041 to 0.976 (mean = 0.595, SD = 0.178). Its distribution shows meaningful variation across institutions, making it well-suited for linear regression modeling.

2.3 Candidate Predictors

The following variables were considered as potential predictors:

Variable	Label	Notes
gradrate	Graduation rate (outcome)	Completers / cohort total
cohort_total	Incoming cohort size	Continuous; filter ≥ 200
tuition2	In-state tuition	Continuous
fee2	In-state required fees	Continuous
avg_faculty_salary	Avg faculty salary	Salary outlays / staff count; 215 missing
roomcap	Dormitory capacity	Dropped: corr = 0.85 with cohort_total
fee3	Out-of-state fees	Dropped corr = 0.86 with fee2
control	Control of institution	3 levels: Public, Private NP, Private FP
hloffer_collapse	Highest level of offering	Collapsed to 4 levels from original 5
locale_collapse	Degree of urbanization	Collapsed from 12 to 4 levels
c21enprf_collapse	Carnegie enrollment profile	Collapsed from 6 to 5 levels
cbsatype	Metro/Micro area type	3 levels; -2 recoded to 3 (Not applicable)
room	On-campus housing offered	Binary Yes/No
board_collapse	Meal plan offered	Collapsed to binary Yes/No
iclevel	Level of institution	Dropped: zero variation (all 4-year)
instcat	Institutional category	Dropped 98.5% one category

Variables roomcap and fee3 were excluded due to high multicollinearity with cohort_total (r = 0.85) and fee2 (r = 0.86) respectively. Variables iclevel and instcat were excluded due to near-zero variance. Room and board amounts (roomamt, boardamt) were excluded due to high missingness (>250 observations) attributable to schools without on-campus housing.

3. Model Selection

3.1 Approach

Model selection was conducted using PROC GLMSELECT in SAS Viya, which handles categorical variables natively via the CLASS statement, eliminating the need for manual dummy coding. Three selection methods (forward, backward, and stepwise) were applied under three criteria (AICc, SBC/BIC, and Adjusted R^2), yielding nine total model selection runs. The candidate predictor set consisted of four continuous variables and seven categorical variables.

3.2 Model Selection Results

Method	Criterion	Effects	R^2	Adj R^2	AICc
Forward	AICc	10	0.7058	0.7006	-3827.93
Backward	AICc	10	0.7058	0.7006	-3827.93
Stepwise	AICc	10	0.7058	0.7006	-3827.93
Forward	Adj R^2	10	0.7058	0.7006	-3827.93
Backward	Adj R^2	10	0.7058	0.7006	-3827.93
Stepwise	Adj R^2	10	0.7058	0.7006	-3827.93
Forward	SBC/BIC	6	0.6938	0.6917	-3809.01
Backward	SBC/BIC	6	0.6938	0.6917	-3809.01
Stepwise	SBC/BIC	6	0.6938	0.6917	-3809.01

Six of nine selection runs: all AICc and Adjusted R2 runs consistently selected the same 10-effect model. The three SBC/BIC runs selected a leaner 6-effect model. This strong convergence across methods and criteria for the larger model provides compelling evidence in its favor.

3.3 Final Model Selection Rationale

The 10-effect AICc/Adj R^2 model was selected as the final model for the following reasons:

- 6 of 9 selection runs agreed on this model across two independent criteria (AICc and Adj R^2)
- Higher explanatory power: $R^2 = 0.7058$ vs. 0.6938 for the SBC model
- The additional predictors (locale, hloffer, c21enprf, board) are theoretically meaningful and contribute to a richer interpretation
- SBC imposes a heavier complexity penalty that may be overly conservative for this sample size

4. Final Model

4.1 Model Fit Statistics

Statistic	Value
Observations Used	1,037
R^2	0.7058
Adjusted R^2	0.7006
Root MSE	0.09479
AIC	-3828.76

AICc	-3827.93
SBC	-4773.82
F Value (df=18, 1018)	135.69
Pr > F	< .0001

The final model explains approximately 70.6% of the variation in baccalaureate graduation rates ($R^2 = 0.7058$), with an adjusted R^2 of 0.7006, indicating the model is not overfitted. The overall F-test is highly significant ($F = 135.69$, $p < .0001$), confirming that the predictors collectively explain a meaningful portion of variation in graduation rates.

4.2 Parameter Estimates

Parameter	Estimate	Std Error	t Value	p-value
Intercept	0.1157	0.0291	3.98	<.0001
control: Private for-profit	-0.2259	0.0560	-4.03	<.0001
control: Private not-for-profit	-0.0268	0.0143	-1.88	0.0606
control: Public (ref)	Reference			
hloffer_collapse 1 (Bach./Postbacc)	-0.0149	0.0224	-0.67	0.5045
hloffer_collapse 2 (Master's)	-0.0277	0.0093	-2.97	0.0031
hloffer_collapse 3 (Post-master's)	-0.0020	0.0122	-0.16	0.8734
hloffer_collapse 4 (Doctoral, ref)	Reference			
locale_collapse 1 (City)	0.0449	0.0157	2.85	0.0044
locale_collapse 2 (Suburb)	0.0566	0.0163	3.47	0.0005
locale_collapse 3 (Town)	0.0482	0.0161	3.00	0.0028
locale_collapse 4 (Rural, ref)	Reference			
c21enprf_collapse 1 (Excl. undergrad)	0.0690	0.0279	2.47	0.0136
c21enprf_collapse 2 (Very high UG)	0.0714	0.0201	3.55	0.0004
c21enprf_collapse 3 (High UG)	0.0495	0.0191	2.59	0.0097
c21enprf_collapse 4 (Majority UG)	0.0216	0.0192	1.13	0.2603
c21enprf_collapse 5 (Majority grad, ref)	Reference			
room: No	-0.0475	0.0311	-1.53	0.1275
room: Yes (ref)	Reference			

board_collapse: No (0)	-0.0379	0.0255	-1.49	0.1378
board_collapse: Yes (1, ref)	Reference			
tuition2	0.0000066	0.000000415	15.90	<.0001
fee2	0.0000085	0.00000246	3.47	0.0005
avg_faculty_salary	0.0000024	0.000000199	11.99	<.0001
cohort_total	0.0000333	0.00000282	11.81	<.0001

4.3 Interpretation of Effects

Continuous Predictors

In-state tuition (tuition2): For each additional dollar of in-state tuition, the predicted graduation rate increases by approximately 0.0000066 ($t = 15.90$, $p < .0001$). While small in magnitude per dollar, this is highly significant. Higher tuition institutions tend to be more selective and better resourced, which is associated with higher graduation rates.

In-state fees (fee2): Each additional dollar in required fees is associated with a 0.0000085 increase in graduation rate ($t = 3.47$, $p = 0.0005$). Similar to tuition, fees serve as a proxy for institutional resources.

Average faculty salary (avg_faculty_salary): Each additional dollar in average faculty salary is associated with a 0.0000024 increase in graduation rate ($t = 11.99$, $p < .0001$). Higher faculty salaries reflect greater institutional investment in instructional quality, which supports student success.

Cohort size (cohort_total): Each additional student in the incoming cohort is associated with a 0.0000333 increase in graduation rate ($t = 11.81$, $p < .0001$). Larger institutions may have more robust student support services and resources that contribute to higher completion rates.

Institutional Control

Public institutions serve as the reference category. Private for-profit institutions have graduation rates that are on average 22.6 percentage points lower than public institutions ($p < .0001$), a large and highly significant effect. Private not-for-profit institutions have rates approximately 2.7 percentage points lower than public institutions, though this effect is only marginally significant ($p = 0.0606$).

Highest Level of Offering (hloffer_collapse)

Doctoral-granting institutions serve as the reference category. Master's-granting institutions (level 2) have graduation rates approximately 2.8 percentage points lower than doctoral institutions ($p = 0.0031$). Bachelor's/postbaccalaureate and post-master's certificate institutions do not differ significantly from doctoral institutions.

Locale (locale_collapse)

Rural institutions serve as the reference category. City (4.5 pp, p = 0.0044), suburban (5.7 pp, p = 0.0005), and town (4.8 pp, p = 0.0028) institutions all have significantly higher graduation rates than rural institutions. Suburban institutions show the largest advantage over rural settings.

Carnegie Enrollment Profile (c21enprf_collapse)

Majority graduate institutions serve as the reference category. Exclusively undergraduate (6.9 pp, p = 0.0136), very high undergraduate (7.1 pp, p = 0.0004), and high undergraduate (4.9 pp, p = 0.0097) institutions all have significantly higher graduation rates. The majority undergraduate category (level 4) does not differ significantly from the reference (p = 0.2603).

Room and Board

Institutions that do not offer on-campus housing have graduation rates approximately 4.8 percentage points lower than those that do, though this effect is not statistically significant (p = 0.1275). Similarly, institutions without a meal plan show rates approximately 3.8 percentage points lower, also not significant (p = 0.1378). Both variables were retained in the model as their removal worsened AICc.

5. Discussion

The final model demonstrates strong predictive performance, explaining over 70% of the variation in baccalaureate graduation rates. The most influential predictors, entered first by all stepwise and forward selection runs, were average faculty salary, in-state tuition, and cohort size, underscoring the importance of institutional resources and scale in predicting student completion.

The strong negative effect of private for-profit status is noteworthy and aligns with well-documented patterns in higher education research. The locale effects suggest that geographic context matters, with rural institutions consistently showing lower graduation rates even after controlling for other institutional characteristics.

Some variables included in the final model have non-significant individual parameter estimates (e.g., room, board_collapse, certain levels of hloffer_collapse and c21enprf_collapse). These were retained because the overall effect contributed to model fit as measured by AICc and Adjusted R2, and because removing them worsened model performance. This is a known feature of categorical variable selection where individual levels may not all be significant even when the overall factor improves the model.

One limitation of this model is that it uses only 1,037 of the 1,252 eligible institutions due to missing values in avg_faculty_salary. Future work could explore imputation strategies to recover

these observations. Additionally, the model is cross-sectional and cannot establish causal relationships between the predictors and graduation rates.